

OSTROLEKA, Poland

WMO: 122850

Lat: 53.066N

Lon: 21.534E

Elev: 94

StdP: 100.20

Time Zone: 1.00 (EUC)

Period: 00-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.6	-12.4	-18.3	0.8	-14.4	-15.6	1.0	-11.5	8.8	5.7	7.9	4.2	1.8	70	0.515

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	30.1	20.4	28.3	19.4	26.8	18.6	21.7	27.8	20.6	26.5	19.6	24.9	3.0	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.6	14.5	24.4	18.6	13.6	23.0	17.6	12.7	22.0	63.8	27.7	59.6	26.5	56.1	25.0	28.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
7.0	6.2	5.4	DB	-18.7	33.9	4.6	1.6	-22.1	35.1	-24.8	36.0	-27.4	36.9	-30.7	38.1	
			WB	-18.0	23.6	5.3	1.0	-21.8	24.3	-24.9	24.8	-27.9	25.4	-31.8	26.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	8.6	-2.1	-1.0	2.8	8.4	13.7	16.7	19.0	18.4	13.6	8.3	4.0	0.7	(6)
(7)		DBStd	8.45	5.31	5.43	4.56	4.07	4.29	3.61	3.30	3.32	3.36	4.05	3.72	4.66	(7)
(8)		HDD10.0	1560	376	307	224	76	15	1	0	0	8	82	183	289	(8)
(9)		HDD18.3	3685	634	540	481	297	153	72	33	41	146	311	431	548	(9)
(10)		CDD10.0	1052	0	0	1	29	130	203	279	262	116	29	2	0	(10)
(11)		CDD18.3	134	0	0	0	0	9	24	53	44	4	0	0	0	(11)
(12)		CDH23.3	1451	0	0	0	9	128	299	508	454	53	0	0	0	(12)
(13)		CDH26.7	381	0	0	0	1	16	74	139	140	11	0	0	0	(13)
(14)	Wind	WSAvg	2.6	3.1	2.9	3.0	2.6	2.5	2.5	2.4	2.2	2.3	2.6	2.8	3.0	(14)
(15)	Precipitation	PrecAvg	630	37	30	41	47	62	82	83	72	67	48	44	39	(15)
(16)		PrecMax	1081	109	58	123	96	128	211	219	250	191	117	87	70	(16)
(17)		PrecMin	456	6	4	8	6	25	26	17	9	10	4	6	10	(17)
(18)		PrecStd	148	23	12	26	24	24	46	46	47	51	29	19	14	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.1	10.3	16.7	24.1	28.6	32.0	32.4	33.3	28.1	21.3	14.2	10.6	(19)
(20)		MCWB	7.1	7.2	10.1	14.9	18.1	22.8	20.3	20.6	19.1	15.7	11.9	8.9		(20)
(21)		2%	DB	7.4	8.6	14.1	21.1	26.6	28.6	29.9	29.9	24.6	18.8	11.9	9.1	(21)
(22)		MCWB	6.0	6.7	8.3	12.9	17.4	19.5	20.1	20.3	17.5	14.4	10.1	7.7		(22)
(23)		5%	DB	6.1	7.0	11.7	18.8	24.6	26.8	28.1	27.8	22.3	16.5	10.3	7.7	(23)
(24)		MCWB	4.9	5.3	7.2	11.4	16.3	18.7	19.5	19.3	16.3	13.3	8.8	6.5		(24)
(25)		10%	DB	4.5	5.5	9.6	16.5	22.2	24.8	26.3	25.7	20.3	14.6	9.0	6.3	(25)
(26)		MCWB	3.5	4.0	6.4	10.2	15.0	17.7	18.8	18.6	15.1	12.2	7.8	5.4		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.6	8.0	11.0	16.0	19.7	24.7	23.0	22.7	19.8	16.6	12.3	9.5	(27)
(28)		MDCB	8.4	9.4	15.3	22.6	26.1	28.8	28.6	29.9	26.3	20.1	14.0	10.3		(28)
(29)		2%	WB	6.2	6.9	9.2	13.7	18.4	21.5	21.5	21.4	18.3	15.1	10.2	7.9	(29)
(30)		MDCB	7.3	8.4	12.5	19.7	24.6	27.2	27.3	28.1	23.2	17.7	11.5	8.8		(30)
(31)		5%	WB	5.0	5.4	7.9	12.3	17.2	19.6	20.5	20.2	17.0	13.7	9.1	6.7	(31)
(32)		MDCB	6.1	7.0	10.8	17.5	23.1	24.8	26.1	26.0	20.9	16.1	10.1	7.6		(32)
(33)		10%	WB	3.6	4.1	6.5	10.9	15.9	18.2	19.5	19.2	15.8	12.4	7.9	5.5	(33)
(34)		MDCB	4.6	5.4	9.3	14.9	21.0	23.0	24.7	24.3	19.5	14.4	8.9	6.4		(34)
(35)	Mean Daily Temperature Range	MDBR	4.4	5.7	8.3	11.2	11.9	11.7	11.0	11.6	10.5	8.3	4.7	4.0		(35)
(36)		5% DB	MDCDBR	4.9	7.0	12.4	15.1	14.7	14.7	13.5	14.3	13.8	11.4	6.6	4.7	(36)
(37)		MCWBR	4.2	5.3	7.7	8.3	6.8	6.5	5.5	6.1	7.1	7.3	5.0	4.0	(37)	
(38)		5% WB	MCDDBR	4.8	6.7	10.9	13.8	12.6	12.6	11.8	12.8	12.1	10.4	6.2	4.6	(38)
(39)		MCWBR	4.1	5.3	7.4	8.0	6.1	6.6	5.3	5.9	6.6	7.0	5.0	4.1		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.308	0.339	0.373	0.420	0.407	0.403	0.436	0.430	0.388	0.362	0.351	0.307		(40)
(41)		taud	2.218	2.220	2.229	2.161	2.260	2.300	2.226	2.237	2.327	2.380	2.286	2.262		(41)
(42)		Ebn at Noon	650	748	802	807	842	849	809	790	781	721	594	580		(42)
(43)		Edh at Noon	66	90	109	133	127	123	131	122	99	77	63	53		(43)
(44)	All-Sky Solar Radiation	RadAvg	0.65	1.36	2.59	4.11	5.24	5.50	5.11	4.43	3.13	1.69	0.66	0.44		(44)
(45)		RadStd	0.12	0.23	0.40	0.55	0.51	0.49	0.58	0.40	0.45	0.29	0.07	0.06		(45)

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(46)
(47)	Regional (20 neighbors)	+0.46	N/A	N/A	N/A	N/A	N/A	-154	+82	N/A	(47)

Nomenclature: See separate page